Peter Polner

3/4/2024

ENVS100

Portfolio

Portfolio

Table of Contents

Introduction 2
Assignments 3
Quiz 2 Reflection 3
Discussion 1: Climate Change & Hope 4
Particulate Graphing Assignment 6
Discussion 2: Indigenous Environmental Justice 11
Supplemental Element 12
Conclusion 14

Introduction

Welcome to my Portfolio, this is a document which outlines my knowledge and thinking that I demonstrated within the Introduction to Environmental Science course (ENVS& 100) under Dr. McKinley at Bellevue College. I chose an environmental science class due to the environment's impact and prevalence within a modern world, relating even to the world of business and economics. Understanding its vastness and capability can help the world avert a climate crisis and prepare for a future with one, understanding our impact on the environment and the environment's impact on us is key to understanding the world around us. Knowing the components of our environment can better help us humans diagnose natural problems effecting our daily life.

Throughout the course I came around to understanding how concepts regarding environment presented in modern media and how that is viewed by scholars, and the importance of knowing primary and secondary sources of information. Expanding on concepts I'm already familiar with created a well-rounded picture of the environment around me. This learning was achieved through a variety of discussions, reflections, and writing assignments, creating a classroom environment where differing opinions could be shared and discussed. The course also tested scientific comprehension, either through having us graph our own discoveries within air quality monitoring or annotating graphs that supplement course materials and modules.

This will be Shown throughout this portfolio, a collection of my best work throughout the course, which will demonstrate learning and growth throughout its content through deep and meaningful reflections. The visual of growth will help track my progress towards meeting the course outcomes, mainly understanding environmental science concepts, and demonstrating that I could make a meaningful member of the scientific community in the future. Understanding the concepts in this course will allow me to grasp the environments effects globally and on differing industries, as well as its effect on people and how they live throughout the world. I hope that you will find this information useful in demonstrating my growth throughout this course.

Peter Polner

Student at Bellevue College

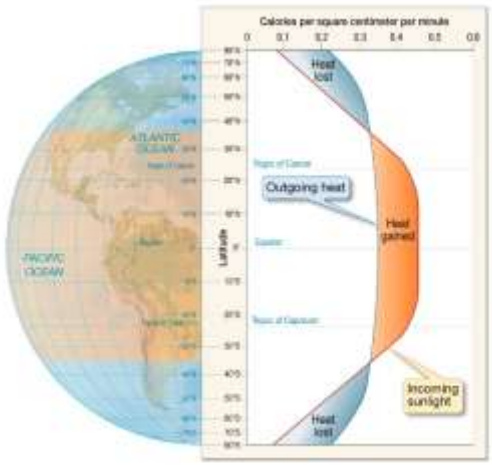
AAS Business (DTA)

Assignments

Quiz 2 Reflection

1. **Learning outcome(s)/Goal(s) within Quiz 2**
    - Outline and describe the major components of an ecosystem.
    - Build skills that empower students to become effective learners and citizens,

2. **Copy your answers to the quiz question or assignment, as well as the related graph or figure if there is one.**

 1) Annotate or explain this figure in detail. Some things you will want to include or consider: What are the axis? What are the units? What do the lines represent? 2) What is the information the figure is trying to teach you? Why is the globe also included in the figure? 3) Explain why heat is gained in some areas of the globe and not others. 4) How does the reflection of solar radiation from Earth's surface add to the effects of uneven solar heating in creating a pole-to-equator heat imbalance? 5) How do you think this diagram would change if all of the ice in the polar regions melted? I am asking you to make a prediction here, based on what you know, you need to respond,	**Answer:** 1. figure shows heat distribution throughout the world, on the Y-axis it shows the latitude of the world (in degrees N/S from equator) and the X-axis shows the calories/cm^2 per Minute 2. the figure tells you that the equator is getting more incoming sunlight, while the northern/southern portions are giving outgoing heat. the map serves as a visual to connect the viewer to their knowledge of the world, and where the uneven heating takes effect. 3. Because the earth spins on its axis, the northern and southern parts receive inconsistent heat, while the equator always gets sunlight and thus heat whichever direction the earth is on its axis, because the earth is a globe. 4. When sun reaches the ice sheets of earth's poles, the light and heat get reflected (albedo effect) creating some greenhouse effect, where radiation is trapped, but that is counteracted, through clouds, and brought to the equator.	**Corrections:** 1. N/A 2. Geography 3. N/A 4. Albedo Effect 5. N/A

	5. If the polar ice melted i thing the heat distributed would look more "normal" on a graph instead of having those weird bumps, because the heat would be evenly distributed throughout the water	

3. **Make corrections and updates with any feedback you received about your work make sure the corrections are clearly highlighted.**
   Corrections are highlighted in red.

4. **Write an explanation of how it displays you have achieved the learning outcome you selected.**
   The sun and its heating of the earth has a ginormous effect on the earth's ecosystem and the workings of its components; like later where I referred to albedo and greenhouse effects effecting polar ice caps. This assignment also created necessary skills to become learners and citizens displaying graph comprehension and the ability to extract data. Understanding these graphs creates effective thinking, that can be used in the real world. Understanding both of these in the term of the graph help us create a better idea of the heating of the earth and its effects.

5. **Even if you do not have corrections to make write a short reflection about what you learned from the assignment and any new information you have built since the assignment was submitted.**
   I further explored previous knowledge in this assignment, by expanding on my knowledge on the rotation of earth, and in the sun's role in heating it, and the many effects that follow that. The assignments focus on the sun and incorporating it with the large example of polar ice helped illuminate climate change effects: playing a role in reflection radiation from the sun, but also melting because of it (due to the followed greenhouse effects). It helped me learn new terms such as the albedo effect. Overall, the assignment created new perspectives on how different sciences interact and how that can be represented through graphs and data.

Discussion 1: Climate Change & Hope

1. **Learning outcome(s)/Goal(s) within Discussion Reflection 1 (Climate Change & Hope)**
   - Build skills that empower you to become effective learners and citizens, including effective notetaking, reading graphs, and using proper citations.
   - Develop an understanding of the nature of science as a human endeavor, with a focus on recognizing and addressing biases within scientific processes, as well as

identifying bias in research articles, news reports, and popular science communication.

- Describe, and give examples of, how each human has an impact on environmental sustainability.

2. Copy your answers to the quiz question or assignment.

Reflection of Warm Regards Podcast: Apocalyptic Narratives with Zeke Hausfather

1) A summary of the 2-3 key ideas or takeaways from the article

1. There are more fears about global warming than needed, this is due to the media landscape and our inability to look past the present (hard to get messaging right).
2. The change between different models becomes less the more CO2 is involved.
3. Groups that have contributed to climate change the most, lobby to not be held accountable.

2) 2-3 connections between what we've learned in class, the learning outcomes and the key points of the article.

There were a lot of connections to our learning outcomes 3 & 8 (regarding pollution's effect on global health and human impact on environmental sustainability respectively) it connects human actions and most importantly policy towards climate action which regulates both pollution and sustainability. The transcript also talked about how "earth system(s) … are non linear" (Gill & Laungani, 2020). Providing examples of complex interconnected systems which are hard to model, we have discussed how earths systems are interconnected, resulting in balancing and reinforcing loops. It also discussed how people (not climate experts) see the IPCC report (Gill & Laungani, 2020), which I think relates to some discussion we had in class about understanding graphs and how science is conducted.

3) A reflection on what you have taken away from the article, connecting it to your life or your reaction to it.

A main point in the article outlined that it is hard to convey scientific messages to the masses, due to the complex nature of the topic. This raised the especially important topic of media literacy and environmental awareness, being aware of bias, standards, and information creates a civic and democratic society, which they further outlined could drastically help humanity stay clear of bad warming scenario (Gill & Laungani, 2020). I think the article is useful in demonstrating the view of the scientific community on this issue, which I saw as stark contrast to media messaging that we get mostly in the form of sensational headlines; explaining the numbers, policy, and science of climate change really helped create an educated big picture idea of the issue and its possible solutions.

4) Your response to the article, questions you discussed with your group, questions you still have, something that surprised you

One thing that really surprised me in the article was how it rejected the notion of an apocalyptic scenario, even in the worst cases the interview conveyed that it is possible to recover from the irreversible change that anthropogenic climate change brings. The interview wasn't all science as well, which is surprising, it brought a good

deal of geopolitics along with the science which I thoroughly enjoyed.

Works Cited:

Hausfather, Z. (2020, December 28). Apocalyptic Narratives, Climate Data, and Hope, with Zeke Hausfather and Diego Arguedas Ortiz. (J. Gill, & R. Laungani, Interviewers) Warm Regards. Retrieved from https://ourwarmregards.medium.com/apocalyptic-narratives-climate-data-and-hope-with-zeke-hausfather-and-diego-arguedas-ortiz-8ed6506260d9

3. **Make corrections and updates with any feedback you received about your work make sure the corrections are clearly highlighted.**
   Corrections are in red. I received feedback on my citations, I only correctly cited the direct quote that I provided but left out the citations for article contents which I paraphrased.

4. **Write an explanation of how it displays you have achieved the learning outcome you selected.**
   The reflection highlighted my comprehension of the article overall but showed a gap in my notation skill (through the citations). I remedied this by reviewing my reflection again and making changes where it was necessary, this gave credit to the people who I got the information from as well as context and reference to those reading my reflection (making any of my claims verifiable). Overall, I believe that this has made me a better scientific citizen, providing useful knowledge for future assignments and academic work.

5. **Even if you do not have corrections to make write a short reflection about what you learned from the assignment and any new information you have built since the assignment was submitted.**
   As I've stated within the assignment, I learned a lot about how the scientific community views climate change, and how that differs from the modern media landscape. I think it helped greatly show how to view bias and look for facts in information that we receive, forming critical thinking. The value of looking at these sources of information helped sharpen my research capabilities, while enabling me to use information from this article (and future ones) to help me learn concepts and look at facts from my own point of view. Overall reflecting on this article helped me grow as a learner in this class and created understanding of how science functions in our modern world.

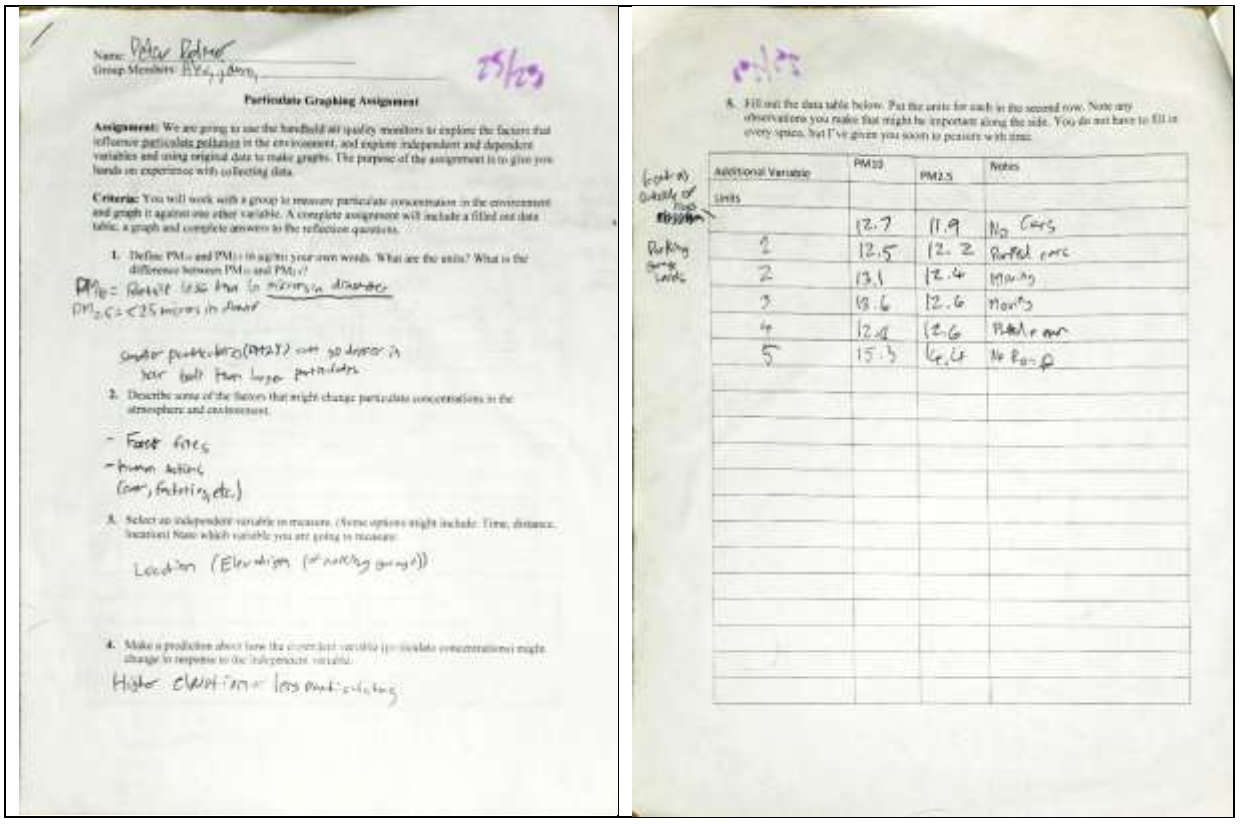
Particulate Graphing Assignment

1. **Learning outcome(s)/Goal(s) within Particulate graphing assignment**

- Build skills that empower you to become effective learners and citizens, including effective notetaking, reading graphs, and using proper citations.

- List the major types of air and water pollution and describe their effects on global health.

2. Copy of answers from assignment



Name: [illegible]
Group Members: [illegible]

Particulate Graphing Assignment

Assignment: We are going to use the handheld air quality monitors to explore the factors that influence particulate pollution in the environment, and explore independent and dependent variables and using original data to make graphs. The purpose of the assignment is to give you hands on experience with collecting data.

Criteria: You will work with a group to measure particulate concentration in the environment and graph it against one other variable. A complete assignment will include a filled out data table, a graph and complete answers to the reflection questions.

1. Define PM_{10} and $PM_{2.5}$ in your own words. What are the units? What is the difference between PM_{10} and $PM_{2.5}$?

PM_{10}: Particle size that is microscopic diameter
$PM_{2.5}$: <2.5 microns in diameter

smaller particulates (PM2.5) can go deeper in your body than larger particulates

2. Describe some of the factors that might change particulate concentrations in the atmosphere and environment.

- Forest fires
- human activity (cars, factories, etc.)

3. Select an independent variable to measure. (Some options might include: Time, distance, location) State which variable you are going to measure:

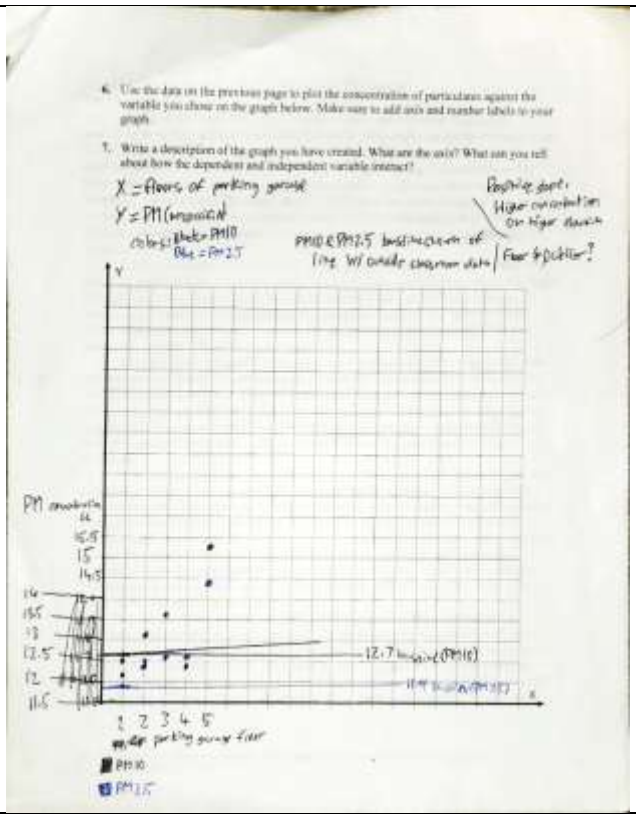
Location (Elevation (# parking garage))

4. Make a prediction about how the [illegible] particulate concentrations) might change in response to the independent variable.

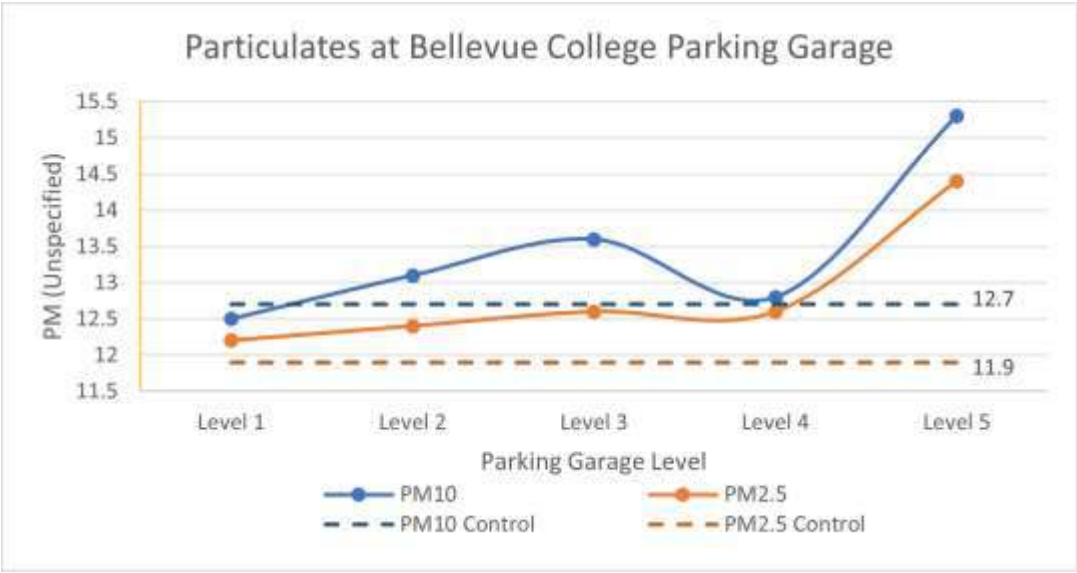
Higher elevation = less particulates

8. Fill out the data table below. Put the units for each in the second row. Note any observations you make that might be important along the side. You do not have to fill in every space, but I've given you room to practice with time.

Additional Variable	PM10	PM2.5	Notes
Units			
	12.7	11.9	No Cars
1	12.5	12.2	Parked cars
2	13.1	12.4	Moving
3	13.6	12.6	Moving
4	12.9	12.6	Parked cars
5	15.3	14.4	No Parking



3. **Make corrections and updates with any feedback you received about your work make sure the corrections are clearly highlighted.**
   As you can see on the last page where I have to graph my data collection, the graph looks very rough, and the many mistakes in pen are visible. To fix this I re-graphed my data in excel!



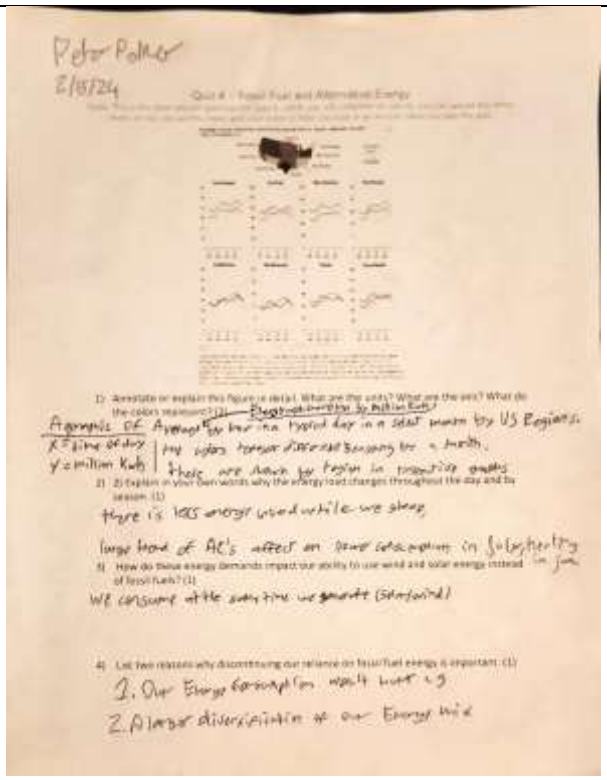
4. **Write an explanation of how it displays you have achieved the learning outcome you selected.**
   I think this further expanded upon the goal of graph literacy to the creation of complex graph that display data to be interpreted scientifically. We also observed particulate pollution within our own community showing how issues such as air pollution can happen close to home.

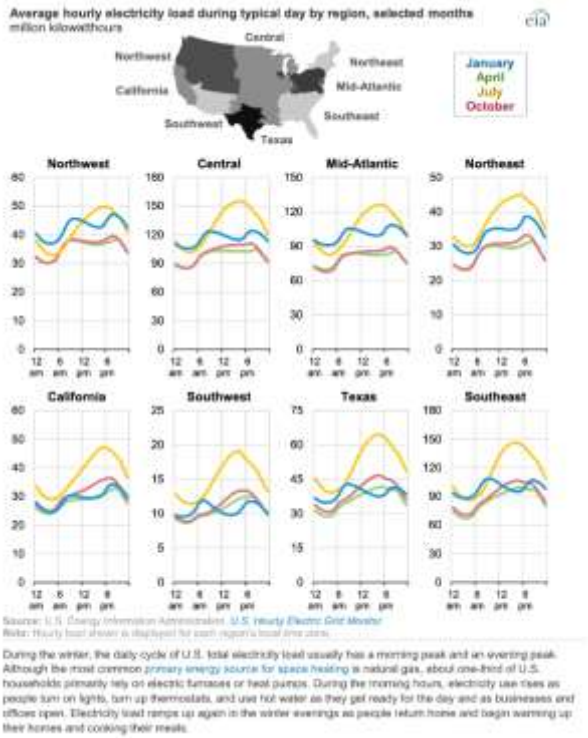
5. **Even if you do not have corrections to make write a short reflection about what you learned from the assignment and any new information you have built since the assignment was submitted.**
   I was very surprised by this assignment, as you see within our graph Levels four and five are noticeable discrepancies, which leaves me to wonder whether it was an outlier or a calibration/human error. I learned even more that science is just a discovery of the unknown guided by the scientific method; if someone else were to look at my graph and say “Hey, that looks funky!” then they can go out and replicate the experiments and confirm or reject my results.

Quiz 4 Reflection

1. **Learning outcome(s)/Goal(s) within Quiz 4**
   - Compare and contrast current conventional energy uses with those of sustainable energy.
   - Describe the ways that human activities have unbalanced biogeochemical cycles, leading to current environmental dilemmas.

2. **Copy of answers from assignment**





3. **Make corrections and updates with any feedback you received about your work make sure the corrections are clearly highlighted.**
   N/A

4. **Write an explanation of how it displays you have achieved the learning outcome you selected.**
   The graph shows the energy use by humans which show how our demand being during the day makes the viability of renewable energy as humanity's primary energy source. Due to our overconsumption and unsmart use of energy (during extreme seasons) have drastic effects on contributions to our environmental problems.

5. **Even if you do not have corrections to make write a short reflection about what you learned from the assignment and any new information you have built since the assignment was submitted.**
   This assignment put into perspective our energy use during the summer and winter for heating and cooling. As well as providing insights on the shortfalls of renewable energy, making me think about ways to mitigate those problems, like constant renewable energy (hydroelectric) and effective energy (nuclear).

Discussion 2: Indigenous Environmental Justice

1. **Learning outcome(s)/Goal(s) within Discussion 2: Indigenous Environmental Justice**
    - Explain how the dumping of solid, toxic, and hazardous wastes relates to environmental health.
    - Compare and contrast current conventional energy uses with those of sustainable energy.

2. **Copy of answers from assignment**

Reflection of Minnesota line 3 pipeline

1) A summary of the 2-3 key ideas or takeaways from the article
    1. Line 3 pipeline goes through ecologically sensitive areas
    2. Disrupts food supply of native peoples (through "wild rice" grain)
    3. Pipeline's process disrupts local watersheds.

2) 2-3 connections between what we've learned in class, the learning outcomes and the key points of the article.

The article connects heavily with our learning outcomes of Explain how the dumping of solid, toxic, and hazardous wastes relates to environmental health & list the major types of air and water pollution and describe their effects on global health. Even though not much real waste or pollution was outlined in the article it did describe on the effect of transporting the hazardous material (tar sand) and its negative effect towards the community. Accidents from the pipeline could immediately destroy local ecology and watersheds (Mercado, 2021), this only illustrates how fossil fuel production can have point-source pollution impacting local populations.

3) A reflection on what you have taken away from the article, connecting it to your life, your reaction to it, or what surprised you.

The article demonstrated the human need for energy consumption and its cost of other local communities. Thankfully there is no proposed pipeline that I know of that would disrupt my life, here in Washington state; however, the energy source of the Snoqualmie river dam directly affects me, living in the Snoqualmie valley. If either the water source of the Snoqualmie river would be contaminated (from a theoretical pipeline) the water I rely on would be unreliable. Or if demand for energy from the dam would decrease causing its dismantling, my home would turn into an island and many nearby communities would be underwater due to being located in floodplains. This made me think about energy demand's effect on my life and the importance of reevaluating out energy mix.

4) Your response to the article, questions you discussed with your group, questions you still have, something that surprised you.

The article I think ended on an open note, without clear resolution so that leads me to wonder: did the lawsuit to stop the pipeline eventually go through? Because I think that

would serve some justice to the people that would've been negatively affected by the pipeline. I'm also interested in differing solutions to this problem, could the pipeline be diverted, or effectively phased out, if all negative effects are mitigated then energy could most definitely be used to support the community instead of destroying it. The article was short and sweet and illustrated major points impacting native communities, while missing out on other points.

Works Cited:

Mercado, A. (2021, October 12). *Minnesota's Line 3 pipeline threatens one of North America's only native grains*. Retrieved from Popular Science: https://www.popsci.com/environment/line-3-indigenous-wild-rice/

3. **Make corrections and updates with any feedback you received about your work make sure the corrections are clearly highlighted.**
   Corrections are in red. I received feedback on my citations, and I believe that my conclusion could have been ended better.

4. **Write an explanation of how it displays you have achieved the learning outcome you selected.**
   Throughout the reflection it referenced the article, and I connected it personally in a correlating way (Energy use)

5. **Even if you do not have corrections to make write a short reflection about what you learned from the assignment and any new information you have built since the assignment was submitted.**
   This assignment created an understanding of how our consumption affects people around us, this is an issue that is ever prevalent within the United States. Our need for energy, things, and food drive us to harness nature to a level that it cannot handle causing us to hoard what we have, leaving many without a way out of poverty, hunger, and similar situations.

Supplemental Element

1. **Learning outcome from supplemental element**
   - Define biodiversity and outline the major threats to biodiversity existing today.
   - Compare and contrast the population dynamics controlling the growth of other species with that of the growing global human population.

2. **Supplemental element**

	
A man made “dam” on the sammamish river	A beaver lodge and eagle nest near the mouth of the sammamish river

3. **Write an explanation of how it displays you have achieved the learning outcome you selected.**
   I went on a walk in Marymoor Park following the Sammamish River and its mouth towards lake Sammamish, there were a couple of things that caught my eye: first a human made damn to help flooding throughout the slough. it’s a major disruption in a working system creating a physical barrier between the same river to improve mostly human interests (considering its near a major population area, Redmond) this is a prime example of how we humans created a thing to help us, while it may be disruptive of wetland species. When I approached the lake, I saw something even more significant, I saw a beaver lodge (the log pile) and an eagle nest, both are species which contribute greatly to biodiversity. They are both specialists, and keystone species contributing to the wellbeing of their environment. In the past bald eagles and other birds were threatened by DDT, a dangerous chemical; but the beaver has a larger problem near the Sammamish River since it cannot build a dam due to the frequent recreational use of the Sammamish River and an artificial dam already in place to regulate water levels.

4. **Short reflection of learning from supplemental element**
   This has brought to my attention how real and present the environment is and how the issues we discussed in class are right in front of us. The places we live, breathe, and consume are affecting species which can mutually benefit us, even just the presence of a

bald eagle nest highlights the importance and effectiveness of conservation practices and conservation ecology. This was a calling to appreciate your surroundings and think about how your actions directly impact your environment.

Conclusion

Throughout this class I had the opportunity to dive into what makes the environment and its ecosystems tick and work in unison. My knowledge was expanded on how different aspects of the ecosystem interact for better and worse, whether its keystone species becoming a lifeline for a habitat or human-made pollution tearing it down, the parts of an environment create an equilibrium and paradigm that is hard to change in any extreme. Natures own checks and balances has shown me why us humans are so different and special compared to other species, we can use this specialty of our emotion and irrationality to help other species thrive outside of our own.

When I began this course, I only had a basic knowledge of how our environment functions, but not interacts. However, throughout this course I learnt that the environment is closely relatable to economics, a topic which I understand more than ecological science. Throughout the course I was able to use some business knowledge to help me learn, and in the future, I will be able to use what I learned in this course about environmental science to help improve my business learning. I was able to see the function of a scientific community and its benefits, a key course goal that was heavily applied towards our learning outcomes. The materials demonstrated within this portfolio demonstrate my critical thinking and application of environmental science; the variation in learning outcomes led me to perceive the environment as a multifaceted system, which it is. The courses use of social discussion and experimentation displayed the scientific pursuit of ecology and environmental sciences, while also focusing on an anthropological level unseen in many other sciences. Overall, I hope you enjoyed the review of my works throughout this terrific semester and that my work displays the nature of my learning throughout the course.

Peter Polner

Student at Bellevue College

AAS Business (DTA)